
UNIVERSITY OF CAPE TOWN
Final Examination - 12 December 2022
Investment Decision Theory and Application
ECO4102Z/ECO4122Z

Time: Three(3)hours

Maximum Marks: 100

INSTRUCTIONS TO CANDIDATES

1. You have 10 minutes at the start of the examination in which to read the questions. You are strongly encouraged to use this time for reading only, but notes may be made on this examination paper. You then have three hours to complete the paper.
2. You must not start writing your answers in the booklet until instructed to do so by the invigilator.
3. Mark allocations are shown in brackets [].
4. Attempt all 6 questions, beginning your answer to each full question on a new page.
5. Show all calculations where this is appropriate.
6. Write legibly using blue or black ink.

1. You have been named as investment adviser to a foundation established by Dr. Godfrey Ndlovu with an original contribution consisting entirely of the common stock of GoMedCo. Founded by Dr. Ndlovu, GoMedCo manufactures and markets medical devices invented by the doctor and collects royalties on other patented innovations.

All of the shares that made up the initial contribution to the foundation were sold at a public offering of GoMedCo common stock, and the R15 million proceeds will be delivered to the foundation within the next week. At the same time, Mrs. Ndlovu will receive R15 million in proceeds from the sale of her stock in GoMedCo.

Dr. Ndlovu's purpose in establishing the Ndlovu Foundation was to "offset the effect of inflation on medical school tuition for the maximum number of worthy students." You are preparing for a meeting with the foundation trustees to discuss investment policy and asset allocation.

- (a) Define and give examples that show the differences between an investment objective, an investment constraint, and investment policy. [6 marks]

- (b) Identify and describe an appropriate set of investment objectives and investment constraints for the Ndlovu Foundation. [4 marks]

[Total: 10 marks]

2. An investor makes decisions based on the utility function $U(w) = w - 6w^2$, where w is the investor's wealth in millions of rand (Rm).

- (a) Demonstrate that the investor has both increasing absolute and relative risk aversion. Show all calculations. [4 marks]

The investor has R50,000 to invest over a 1-year period and has no other wealth. They have three options:

- i. Invest in a risk-free account. There will be no change in the value of the investment over 1 year.
- ii. Invest in an asset that will give a 60% return over 1 year with probability 0.2, a 20% return with probability 0.7 and a -40% return with probability 0.1.
- iii. Invest in an asset that will give a 30% return with probability 0.5 and a 20% return with probability 0.5.

The investor makes no allowance for discounting when making investment decisions. The investor must invest the whole R50,000 in a single option.

- (b) Determine which option the investor should choose to maximise their expected utility at the end of the year. [6 marks]

- (c) Comment on why the investor could not use $U(w)$ to choose from the above options if their initial wealth was R65,000. [3 marks]

[Total: 13 marks]

3. Define the following terms:

- (a) Mean-preserving spread [1 mark]
 (b) Second-order stochastic dominance [1 mark]

Consider the following investments whose expected returns(%), have the following probability distribution(s):

Return	Probability	
	Investment 1	Investment 2
10%	0.25	0.20
12%	0.25	0.50
14%	0.25	0.10
15%	0.25	0.20

- (c) Compare the two investments using the second-order stochastic dominance criterion. [5 marks]

[Total: 7 marks]

4. Suppose that there are two risky assets: X and Y , with the following properties.

	Expected Return	Standard Deviation
Asset X	0.07	0.2
Asset Y	0.12	0.4

The risk-free rate is 3 percent, and the correlation between assets X and Y is 0.45.

- (a) Determine the expected return and standard deviation of a portfolio of X and Y that maximizes the Sharpe ratio (i.e. optimal portfolio). Assume no restrictions to short selling. [10 marks]
 (b) Investors can hold any portfolio consisting of asset X and asset Y and also the risk-free asset. In addition, suppose the Satrix Top40 fund offers an expected return of 9% and a standard deviation of 23.9%. All investors have standard mean-variance preferences. Should any investor invest in the Satrix Top40 fund? Explain clearly why or why not. [3 marks]

[Total: 13 marks]

5. (a) Outline the three forms of efficient markets hypothesis (EMH). [3 marks]
 (b) ABC Company has just announced that its profits are up by 42% on last year. On the announcement ABC shares fell in price by 10%. Analysts had been predicting a rise in profits of 55%. A friend says that this shows that the efficient markets hypothesis is false. Comment on this statement. [4 marks]

Abigail is a fund manager at Diversified Portfolio Company(DPC). She is certain that her portfolio is fully diversified with a CAPM beta of 1.0. The risk free rate is 8% and the CAPM risk premium ($E(R_m) - R_f$) is 6.2%. The company's research department has, over the past years, conducted research and have concluded there are two risk factors: changes in industrial production, $\bar{\gamma}_1$ and unexpected inflation $\bar{\gamma}_2$. The APT equation is:

$$E(R_i) - R_f = \beta_{i1}(\bar{\gamma}_1 - R_f) + \beta_{i2}(\bar{\gamma}_2 - R_f)$$

$$E(R_i) = 0.08 + \beta_{i1}0.05 + \beta_{i2}0.11$$

- (c) If her portfolio has a sensitivity factor, $\beta_{i1} = -0.65$ to the first factor, what is its sensitivity to unexpected inflation? [3 marks]
- (d) Abigail is planning to rebalance her portfolio to keep the same expected return but reduce exposure to inflation to zero (i.e. $\beta_{i2} = 0$), what will be its sensitivity to the first factor? [3 marks]

[Total: 13 marks]

6. (a) Suppose, due to recent economic developments in the country, you expect technology stocks, such as software development companies and hardware equipment firms, to perform well over the next three to six months. As an active portfolio manager, describe how would you take advantage of this forecast. [3 marks]
- (b) A charity and an insurance company invest their assets with the same investment manager. The finance directors of the two institutions, who know each other well, are discussing the performance of the investment manager over dinner. They are surprised that the returns of their respective portfolios have differed considerably over the last year. Suggest reasons why the performance of the two portfolios might differ. [4 marks]

Consider the following trading and performance data for four different equity mutual funds:

	Fund 1	Fund 2	Fund 3	Fund 4
Assets under Management, Avg. for Past 12 months (mil)	ZAR289.4	ZAR653.7	ZAR1,298.4	ZAR5,567.3
Security Sales, Past 12 months (mil)	ZAR37.2	ZAR569.3	ZAR1,453.8	ZAR437.1
Expense Ratio	0.33%	0.71%	1.13%	0.21%
Pretax Return, 3-year avg.	9.98%	10.65%	10.12%	9.83%
Tax-adjusted Return, 3-year avg.	9.43%	8.87%	9.34%	9.54%

- (c) State, with reasons, which two funds are most likely to be actively managed and which two are most likely passive funds. [3 marks]

- (d) Determine which funds were the most and least tax efficient in the operations, over the three-year period. [6 marks]

[Total: 16 marks]

7. You are an investment consultant to the trustees of a trust that has used two equity fund managers (Managers A and B) since 1 January 2021. You are given information below about the market value of the trust's funds with Managers A and B. You are also given data on the capital index for their benchmark for 2021. You are further told that in addition to the initial investments made by the trust on 1 January 2021, Managers A and B received a further R100 million and R50 million of new money; respectively, from the trust on 1 July 2021. The dividend yield on the benchmark capital index is 3% p.a. on 31 December 2021.

	Manager A Fund ZAR millions	Manager B Fund ZAR millions	Benchmark Capital index
1 January 2021	200	100	1150
30 June 2021	130	90	920
31 December 2021	400	180	1262

A trustee observes that “Manager A has produced a gain of 33.3% on funds invested (i.e. R100 million on R300 million invested) compared to Manager B’s 20% (i.e. R30 million on R150 million invested”. The trustee suggests that the fund should only use Manager A.

- (a) By calculating returns earned by the fund managers, and the total return for the benchmark, comment on the trustee’s suggestion. [6 marks]

The trustee furnishes you with additional information for risk-adjusted returns to be calculated for Managers A and B. The data is for the three years 2018-2021, and the risk-free rate of return over this period was 6% p.a.

	Manager A	Manager B	Benchmark
Annual return (% p.a.)	12.0	10.5	10.0
Standard deviation (% p.a.)	15.0	7.5	9.5
Correlation coefficient with benchmark	0.75	0.45	1.00
Tracking error (% p.a.)	2.5	0.5	0.0

- (b) Using the above information, calculate four different risk-adjusted performance measures for Managers A and B. [4 marks]
- (c) Comment briefly on the results from part (b), stating any limitations that apply to them. [3 marks]
- (d) Outline the main advantages of the fundamental law of active portfolio management. [3 marks]

[Total: 16 marks]

8. Following the financial crisis in 2008 a number of economists were interviewed and claimed that they had been expecting something like the crisis to occur; however, there was no record of them having expressed this opinion before the crisis.

(a) Describe the behaviour these economists were demonstrating. [4 marks]

An investor has been investing their own money for a period of years despite the fact that a better return could have been produced by investing the money in a risk-free investment that produced a modest income.

(b) State, with justification, any two behaviours that may have led to the investor continuing to invest as had been done in the past. [4 marks]

An investment manager has to sell a number of stocks in the near future. They have decided to sell only those stocks that are trading above the price they paid for them and to make the sales the following Friday before they enjoy the weekend prior to a public holiday the following Monday.

(c) Describe two behaviours that might account for these decisions. [4 marks]

[Total: 12 marks]
